

$$1. \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

rozpíšme:

$$\|\vec{u} + \vec{v}\|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle \stackrel{SL2}{=} \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \|\vec{u}\|^2 + \|\vec{v}\|^2 + (\langle \vec{u}, \vec{v} \rangle + \overbrace{\langle \vec{v}, \vec{u} \rangle}^{SSS})$$

$$= \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

↑ lebo $x + \bar{x} = 2 \operatorname{Re}(x)$

$$\|\vec{u} - \vec{v}\|^2 = \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle \stackrel{SL2}{=} \langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle - \langle \vec{v}, \vec{v} \rangle = \|\vec{u}\|^2 + \|\vec{v}\|^2 - (\langle \vec{u}, \vec{v} \rangle + \overbrace{\langle \vec{v}, \vec{u} \rangle}^{SSS})$$

$$= \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

odčítame:

$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) - \|\vec{u}\|^2 - \|\vec{v}\|^2 + 2 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = 4 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) \quad / : 4$$

$$\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

$$2. \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$

rozpíšme:

$$\|\vec{u} - i\vec{v}\|^2 = \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle \stackrel{SL2}{=} \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -i\vec{v} \rangle + \langle -i\vec{v}, \vec{u} \rangle + \langle -i\vec{v}, -i\vec{v} \rangle \stackrel{SL1}{=} \|\vec{u}\|^2 + \|\vec{v}\|^2 - i \langle \vec{u}, \vec{v} \rangle - i \overbrace{\langle \vec{v}, \vec{u} \rangle}^{SSS}$$

$$= \|\vec{u}\|^2 + \|\vec{v}\|^2 - i \langle \vec{u}, \vec{v} \rangle + i \overbrace{\langle \vec{u}, \vec{v} \rangle}^{SSS} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2i \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

$$\|\vec{u} + i\vec{v}\|^2 = \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle \stackrel{SL2}{=} \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle \stackrel{SL1}{=} \|\vec{u}\|^2 + i \langle \vec{u}, \vec{v} \rangle + \|\vec{v}\|^2 + i \overbrace{\langle \vec{v}, \vec{u} \rangle}^{SSS}$$

$$= \|\vec{u}\|^2 + \|\vec{v}\|^2 + i \langle \vec{u}, \vec{v} \rangle - i \overbrace{\langle \vec{u}, \vec{v} \rangle}^{SSS} = \|\vec{u}\|^2 + \|\vec{v}\|^2 + i \cdot 2i \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

odčítame:

$$\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) - \|\vec{u}\|^2 - \|\vec{v}\|^2 + 2 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

$$\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 = 4 \operatorname{Im}(\langle \vec{u}, i\vec{v} \rangle) \quad / : 4$$

$$\operatorname{Im}(\langle \vec{u}, i\vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$

× keďže $z - \bar{z} = (a+bi) - (a-bi) = 2bi = 2i \operatorname{Im}(z)$